Parities, differences and neighbours: OEIS A239151

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Abstract

OEIS A239151 carries an empirical recurrence of order 14 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is decided by a bounded amount of state carried down the array; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 15$ vertices and is C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A239151 is "Number of (n+1)X(2+1) 0..3 arrays with no element equal to all horizontal neighbors or unequal to all vertical neighbors, and new values 0..3 introduced in row major order.". It has offset 1 and begins

$$2, 2, 55, 208, 2778, 17050, 166531, 1221727, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 4*a(n-1) +51*a(n-2) -84*a(n-3) -529*a(n-4) +442*a(n-5) +1605*a(n-6) +1916*a(n-7) -5423*a(n-8) -5800*a(n-9) +8753*a(n-10) +890*a(n-11) -5088*a(n-12) +4464*a(n-13) +3456*a(n-14)

As of the "Last modified" line on the live entry (Jul 23 2025 10:57:17, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays with 3 columns over the alphabet $\{0, 1, 2, 3\}$, growing one row at a time, and asks that no element may be equal to all of its horizontal neighbours, and none may differ from all of its vertical neighbours.

Every cell the condition names lies within one row of the cell being judged, so three consecutive rows decide everything anchored at the middle one. Carry two consecutive rows as the state; a row is judged when the row after it arrives, and the row with nothing after it is judged by the end vector. A neighbour off the edge of the array is not a neighbour, which is what the entry means and what makes the first and last rows behave differently from the rest.

The entry also asks that new values be introduced in row major order, which is not a condition on any window at all: whether a value may appear here depends on the whole prefix of the array. It enters through a single integer — how many values have been introduced so far — and the effect is that each array is counted once for the whole class of arrays differing from it only by renaming the values.

Merging vertices with identical futures leaves $S = 15$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) \;=\; \iota^{\top} M^{\,n} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^{14} - \sum_i c_i t^{14-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^{j} q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+14} - \sum_i c_i M^{j+14-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. □

Theorem 2.

$$a(n) \;=\; 4\,a(n{-}1) + 51\,a(n{-}2) - 84\,a(n{-}3) - 529\,a(n{-}4) + 442\,a(n{-}5) + 1605\,a(n{-}6) + 1916\,a(n{-}7) - 5423\,a(n{-}8) -$$

for every $n > 13$.

Proof. The vector $w = q(M)\tau$ was formed by 14 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 13 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one and tests the entry's condition on it directly. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A239151.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.